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ELECTRONIC APPARATUS

Abstract

An electronic apparatus includes a fixed unit, a movable unit that holds an image sensor, is opposite to the fixed unit, and is movable relative to the fixed unit, a heat radiating plate opposite to a surface of the fixed unit, which is opposite to a surface of the fixed unit that is opposite to the movable unit, a first heat radiating member that includes a first connector connected to the movable unit, and a second connector connected to a side of the fixed unit, which is opposite to the heat radiating plate, and a second heat radiating member disposed between the second connector of the first heat radiating member and the heat radiating plate. The first heat radiating member and the heat radiating plate are thermally connected via the second heat radiating member.

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to an electronic apparatus.

Description of Related Art

[0002] One conventional electronic apparatus such as an image pickup apparatus has a substrate wiring structure in which a movable unit supported movably on a fixed unit (support unit) and a control substrate are connected by a flexible printed circuit (FPC). For example, an image pickup apparatus having an optical image-stabilizing function of an object achieves image stabilization of the object by moving the movable unit supporting the image sensor relative to the fixed unit in a direction orthogonal to the optical axis.

[0003] Since an image sensor provided on the movable unit is to be cooled, a flexible heat-radiating member such as a graphite sheet is usually provided. Japanese Patent Application Laid-Open No. 2009-284414 discloses an image pickup apparatus in which a flexible heat-radiating member is connected to cool the image sensor while the fixed unit and the movable unit are electrically connected.

[0004] For the image pickup apparatus disclosed in Japanese Patent Application Laid-Open No. 2009-284414, a reaction force generated by deformation of the heat-radiating member becomes a load in driving the movable unit. Thus, the heat-radiating member that is less flexible may hinder the driving of the movable unit. Since the reaction force of the heat-radiating member applies a load in the optical axis direction of the image sensor, the attachment accuracy of the image sensor may decrease.

SUMMARY

[0005] An electronic apparatus according to one aspect of the disclosure includes a fixed unit, a movable unit that holds an image sensor, is opposite to the fixed unit, and is movable relative to the fixed unit, a heat radiating plate opposite to a surface of the fixed unit, which is opposite to a surface of the fixed unit that is opposite to the movable unit, a first heat radiating member that includes a first connector connected to the movable unit, and a second connector connected to a side of the fixed unit, which is opposite to the heat radiating plate, and a second heat radiating member disposed between the second connector of the first heat radiating member and the heat radiating plate. The first heat radiating member and the heat radiating plate are thermally connected via the second heat radiating member.

[0006] An electronic apparatus according to another aspect of the disclosure includes a fixed unit, a movable unit that holds an electronic component and is movable relative to the fixed unit, a first heat radiating member and a second heat radiating member configured to transfer heat from the electronic component, a control substrate configured to transmit a signal, a heat radiating plate provided between the fixed unit and the control substrate, a third heat radiating member configured to transfer heat from the control substrate, and a first connecting member configured to electrically connect the movable unit and the control substrate. The first heat radiating member or the second heat radiating member and the third heat radiating member are connected to the heat radiating plate. The heat radiating plate has an opening. A first length of the opening in a direction orthogonal to a transmission direction of the signal is longer than a second length of the opening in the transmission direction in the first connecting member.

[0007] Further features of various embodiments of the disclosure will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIGS. **1A** and **1B** are perspective views of an electronic apparatus according to this embodiment.

[0009] FIG. **2** is an exploded perspective view of the electronic apparatus according to this embodiment.

[0010] FIGS. **3A** and **3B** are perspective views of an image stabilizing unit according to this embodiment.

[0011] FIGS. **4A** and **4B** are exploded perspective views of the image stabilizing unit according to this embodiment.

[0012] FIGS. **5A** and **5B** are sectional views of the image stabilizing unit according to this embodiment.

[0013] FIGS. **6A** and **6B** are layout diagrams of a heat radiating member provided on a heat radiating plate according to this embodiment.

[0014] FIG. **7** is a schematic view of a control substrate according to this embodiment.

[0015] FIG. **8** is an exploded oblique view of the electronic viewfinder (EVF) unit according to this embodiment.

DETAILED DESCRIPTION

[0016] Referring now to the accompanying drawings, a detailed description will be given of embodiments according to the disclosure.

[0017] This embodiment will discuss an image pickup apparatus having an image sensor (heat generator) as an example of an electronic apparatus, but is not limited to this example and is applicable to another electronic apparatus other than the image pickup apparatus (electronic apparatus having an electronic component (heat generator) other than the image sensor).

[0018] Referring now to FIGS. **1A** and **1B**, a description will be given of a schematic configuration of an image pickup apparatus (electronic apparatus) **10** according to this embodiment. FIGS. **1A** and **1B** are perspective views of the image pickup apparatus **10**. Regarding the direction of the image pickup apparatus **10**, the object side is defined as the front (surface) side based on a direction viewed from the user, and the vertical (up-down) direction, the depth (front-rear) direction, and the lateral (left-right) direction are defined as viewed from the user directly facing the back of the image pickup apparatus **10**. Thus, FIG. **1A** is a perspective view (front perspective view) of the image pickup apparatus **10** viewed from the front side (object side), and FIG. **1B** is a perspective view (rear perspective view) of the image pickup apparatus **10** viewed from the rear side (rear side). This embodiment will discuss a lens interchangeable type camera in which a lens apparatus (interchangeable lens) is attached to and detachable from a camera body as the image pickup apparatus **10**, but is not limited to this example and is applicable to an image pickup apparatus in which a camera body and a lens apparatus are integrated.

[0019] The image pickup apparatus **10** includes an exterior part **10c**. The exterior part **10c** includes a plurality of members. The image pickup apparatus **10** includes a mount **10a** on the front side. The interchangeable lens (lens apparatus) (not illustrated) can be mounted on the mount **10a**. An axis passing through the center of the mount **10a** approximately coincide with an optical axis P (alternate long and short dash line in FIG. **1A**) of the imaging optical system of the interchangeable lens, that is, the imaging optical axis. The image pickup apparatus **10** includes a liquid crystal monitor **71** and an EVF panel **81**, which display image data captured by an image sensor **230** and setting information about imaging.

[0020] Referring now to FIG. **2**, the internal configuration of the image pickup apparatus **10** will be described. FIG. **2** is an exploded perspective view of the main parts of the image pickup apparatus **10** viewed from the rear side (back side). FIG. **2** omits some components of the image pickup

apparatus **10**, such as the exterior part **10c**. FIG. 2 and subsequent figures will illustrate only the portions necessary for understanding this embodiment, and will omit unnecessary portions.

[0021] The image pickup apparatus **10** includes a control substrate **100**, an image stabilizing unit **200**, a shutter unit **300**, a base member **400**, and a heat radiating plate **700**. The image stabilizing unit **200** constitutes an image stabilizing apparatus that performs image stabilization of an image. A control unit of the image stabilizing apparatus is provided on the control substrate **100**.

[0022] The image stabilizing unit **200** includes a movable optical member. The image stabilizing unit **200** is fixed to the base member **400** together with the shutter unit **300**. The image stabilizing unit **200** is held by the base member **400** to which the shutter unit **300** is assembled and fixed. For example, the image stabilizing unit **200** is supported by three screws **600a**, **600b**, and **600c** and three coil springs **500a**, **500b**, and **500c** so as to be displaceable (movable) in a direction (optical axis direction) along the optical axis P (see FIG. 1A) relative to the base member **400**. An assembly worker adjusts fastening amounts of the screws **600a**, **600b**, and **600c**. This structure can adjust the tilt of the imaging surface of the image sensor (electronic component) **230** (see FIG. 4B) relative to the base member **400**, and the position on the optical axis P. Once the adjustment of the tilt of the imaging surface is completed, the screws **600a**, **600b**, and **600c** are adhered to and fixed to the fixed unit of the image stabilizing unit **200** to prevent them from loosening. The fixed unit **200b** (see FIG. 3) is a support member, and its details will be described later with reference to FIG. 3.

[0023] The control substrate **100** and the base member **400** are fixed to the exterior part **10c**. A signal (imaging signal) output from the image sensor **230** is transmitted to the control substrate **100**. A control IC **101** that is used to control the imaging signal, and connectors **102**, **103**, and **104** are mounted on the control substrate **100**. Various electronic components (not illustrated) such as chip resistors, ceramic capacitors, inductors, and transistors are also mounted on the control substrate **100**.

[0024] The image pickup apparatus **10** includes a first connecting member **270a**, a second connecting member **270b**, and a third connecting member **240**, each extending from the image stabilizing unit **200** as flexible wiring members. The first connecting member **270a** is connected to the connector **102**. The second connecting member **270b** is connected to the connector **103**. The third connecting member **240** is connected to the connector **104**. Thereby, the control substrate **100** and the image stabilizing unit **200** are electrically connected.

[0025] The heat radiating plate **700** is disposed between the control substrate **100** and the image stabilizing unit **200** (fixed unit **200b**). The heat radiating plate **700** is supported by earrings **401** and **402**, a bottom heat radiating plate **403**, and a top heat radiating plate **404**, which are plate members attached to the base member **400** together with the control substrate **100** by screws **700a**, **700b**, **700c**, **700d**, and **700e**. Each of the heat radiating plate **700**, the bottom heat radiating plate **403**, and the top heat radiating plate **404** is made of a metal material such as copper or aluminum, which has a high thermal conductivity. The earrings **401** and **402** are made of a metal material such as stainless steel or iron, which has a lower thermal conductivity than that of metal materials such as copper or aluminum, in order to suppress the transfer of heat to the grip portion held by the user of the image pickup apparatus **10**.

[0026] The image stabilizing unit **200** includes a first heat radiating member **801**. The first heat radiating member **801** is connected to second elastic heat radiating members (second heat radiating members) **802a** and **802b** provided on the heat radiating plate **700**, and transfers heat from the image stabilizing unit **200** to the heat radiating plate **700**. Third elastic heat radiating members (third heat radiating members) **803a**, **803b**, and **803c** are provided between the heat radiating plate **700** and the control substrate **100**. The third elastic heat radiating members **803a**, **803b**, and **803c** dissipate (transfer) heat from electronic components such as ICs provided on the control substrate **100** to the heat radiating plate **700**.

[0027] Referring now to FIGS. 3A, 3B, 4A, and 4B, the image stabilizing unit **200** will be described. FIG. 3A is a front perspective view of the image stabilizing unit **200**, and FIG. 3B is a

rear perspective view of the image stabilizing unit **200**. FIG. 4A is an exploded front perspective view of the image stabilizing unit **200**, and FIG. 4B is an exploded rear perspective view of the image stabilizing unit **200**.

[0028] The image stabilizing unit **200** includes a movable unit **200a** and a fixed unit **200b**. The movable unit **200a** is a movable member including the image sensor **230**. The fixed unit **200b** is a support member fixed to the base member **400**. The fixed unit **200b** is configured so that the position of the image sensor **230** can be adjusted in the optical axis direction. The movable unit **200a** is supported by the fixed unit **200b** in a state in which it can be displaced in an arbitrary direction (a direction different from the optical axis direction) in a plane orthogonal to the optical axis P relative to the fixed unit **200b**. The optical image stabilization is achieved by moving the movable unit **200a** in the direction orthogonal to the optical axis P.

[0029] The fixed unit **200b** mainly includes a front yoke **210**, a fixed base plate **250**, and a rear yoke **260**. The movable unit **200a** mainly includes a movable base holder **220**, an image sensor substrate **231**, a sensor holder **280**, and a third connecting member **240**. Each of the first connecting member **270a** and the second connecting member **270b** connects the image sensor substrate **231** and the control substrate **100**. The third connecting member **240** connects the movable base holder **220** and the control substrate **100**. Each of the first connecting member **270a**, the second connecting member **270b**, and the third connecting member **240** includes an FPC.

[0030] The movable unit **200a** includes the image sensor substrate **231**. The image sensor **230** is mounted on the image sensor substrate **231**. The image sensor **230** is a complementary metal oxide semiconductor (CMOS) image sensor or a charge-coupled device (CCD) image sensor, and converts an optical image of an object into an electrical signal. The image sensor **230** and the image sensor substrate **231** are adhered to and fixed to the movable base holder **220**. The sensor holder **280** equipped with an optical low-pass filter **281** is disposed in front of the image sensor **230**. The optical low-pass filter **281** is an optical element for preventing the incidence of infrared rays and preventing the occurrence of color moiré.

[0031] Three openings are formed in the movable base holder **220**, and three coils **241a**, **241b**, and **241c** are mounted in the three openings. The third connecting member **240** is disposed so as to overlap the three coils **241a**, **241b**, and **241c**. One end of the third connecting member **240** is electrically connected to the coils **241a**, **241b**, and **241c**. The other end of the third connecting member **240** is electrically connected to the connector **104** of the control substrate **100**.

[0032] The movable base holder **220** has three metal balls arranged in three ball receivers **220a**, **220b**, and **220c** (see FIG. 4B), which are received on the plane portion of the opposing fixed base plate **250**. When the three metal balls roll, the movable base holder **220** can be displaced in an arbitrary direction in a plane orthogonal to the optical axis P. Magnets **212a**, **212b**, **212c** are adhered to and fixed to the front yoke **210** (see FIGS. 4A and 4B) at predetermined positions on the surface opposite to the movable base holder **220**.

[0033] The fixed base plate **250** has openings **251a**, **251b**, and **251c** formed at different positions when viewed from the direction of the optical axis P (optical axis direction). Magnets **252a**, **252b**, and **252c** are respectively incorporated in the openings **251a**, **251b**, and **251c**. On the fixed base plate **250**, pillars **251a**, **251b**, and **251c** are erected facing the front yoke **210** to fix the front yoke **210**, and the front yoke **210** and the fixed base plate **250** are joined so as to sandwich the movable base holder **220**.

[0034] When viewed from the direction of optical axis P, the magnets **212a**, **212b**, **212c**, **252a**, **252b**, and **252c** are formed in approximately the same positions and shapes as those of corresponding coils **241a**, **241b**, and **241c**, respectively, and sandwich the coils **241a**, **241b**, and **241c**. The magnets **212a**, **212b**, **212c**, **252a**, **252b**, and **252c** are arranged at positions so that their centers are approximately aligned with the centers of the corresponding coils **241a**, **241b**, and **241c**. An electric signal from the control substrate **100** is applied to the coils **241a**, **241b**, and **241c** via the third connecting member **240**. At this time, electromagnetic forces are generated between the

magnets **212a**, **212b**, and **212** and the magnets **252a**, **252b**, and **252c**. This electromagnetic force allows the movable base holder **220** to perform optical image stabilization.

[0035] The assembly worker attaches the rear yoke **260** to the fixed base plate **250** from the rear side so that the magnets **252a**, **252b**, and **252c** are housed inside the openings **251a**, **251b**, and **251c**, respectively. Each of the rear yoke **260** and the fixed base plate **250** is made of a ferromagnetic material. The assembly worker can achieve magnetic attraction by simply aligning the rear yoke **260**, to which the magnets **252a**, **252b**, and **252c** are attached, with the fixed base plate **250** and bringing them into contact. In other words, the two parts can be joined without using adhesive.

[0036] An opening **253** is formed in the fixed base plate **250**. While the movable base holder **220** is sandwiched between the front yoke **210** and the fixed base plate **250**, the image sensor substrate **231** is exposed from the opening **253** to the rear side. The first connecting member **270a** and the second connecting member **270b** are connected to the rear surface of the image sensor substrate **231**. The assemble worker installs the first connecting member **270a** and the second connecting member **270b** into the image sensor substrate **231** from the rear side by passing them through the opening **253**. They are electrically connected by being connected with solder or a connector.

[0037] Each of the first connecting member **270a** and the second connecting member **270b** has a long plate shape in the vertical direction of the image pickup apparatus **10**, is bent in a semicircular arc shape, and is flexible so as not to interfere with the image stabilization of the movable base holder **220**. As described above, the first connecting member **270a** is connected to the connector **102** of the control substrate **100**, and the second connecting member **270b** is connected to the connector **103**. As illustrated in FIG. 3B, the transmission direction of the first connecting member **270a** and the second connecting member **270b** is the longitudinal direction of the first connecting member **270a** and the second connecting member **270b**, which is the vertical direction of the image pickup apparatus **10**. The first connecting member **270a** and the second connecting member **270b** transmit the electric signal from the image sensor substrate **231** to the control substrate **100**.

[0038] Referring now to FIGS. 5A and 5B, a description will be given of a structure according to this embodiment for obtaining a high heat radiating effect while suppressing an increase in load caused by the displacement of heat radiating members such as the first heat radiating member **801**. FIG. 5A is a sectional view of a main part of the image stabilizing unit **200**, and corresponds to a section taken along a line A-A of the image stabilizing unit **200** illustrated in FIG. 3B. The movable base holder **220** is made of a magnesium alloy or aluminum alloy, which is a metal with high heat dissipation property, and transfers heat from the image sensor substrate **231** that is adhered to the movable base holder **220**.

[0039] The first heat radiating member **801** is a flexible sheet member such as a graphite sheet. The center of the first heat radiating member **801** (fixed parts (first connectors) **801a** and **801b**) is adhered to and fixed to the movable base holder **220**. Both ends of the first heat radiating member **801** (fixed parts (second connectors) **801c** and **801d**) are bent into a semicircular arc shape and adhered to and fixed to the fixed holder **290** provided on the fixed base plate **250**. Thereby, the first heat radiating member **801** can reduce the load applied to the movable base holder **220** in the direction along the optical axis P (optical axis direction). Since the first heat radiating member **801** is flexible, when the movable base holder **220** moves in the plane orthogonal to the optical axis P due to the image stabilization, the first heat radiating member **801** deforms, thereby suppressing the load.

[0040] The fixed parts (first connectors) **801a** and **801b** at the center of the first heat radiating member **801** and the fixed parts (second connectors) **801c** and **801d** having semicircular shapes at both ends are arranged so as to approximately coincide in the optical axis direction. That is, when viewed from a direction orthogonal to the moving direction of the movable unit **200a** (the in-plane direction in which the movable unit **200a** moves), at least parts of the fixed parts **801c** and **801d** of the first heat radiating member **801** overlap the fixed parts **801a** and **801b**. This provides uniform

flexibility with respect to the optical axis P in a plane orthogonal to the optical axis P.

[0041] The second elastic heat radiating members **802a** and **802b** provided on the heat radiating plate **700** contact (are pressed against) both end portions (fixing parts **801c** and **801d**) of the first heat radiating member **801** adhered to and fixed to the fixed holder **290** and the heat radiating plate **700**, and radiate heat from the image sensor substrate **231** to the heat radiating plate **700**. The second elastic heat radiating members **802a** and **802b** are elastic members such as heat-dissipating rubber or a graphite sheet rolled into a cylindrical shape that has elasticity (flexibility) at least in the optical axis direction (direction orthogonal to the moving direction of the movable unit **200a**). Thereby, the reaction force against the image stabilizing unit **200** in the optical axis direction can be suppressed, and a decrease in the installation accuracy of the image stabilizing unit **200** in the optical axis direction can be suppressed. The load in the optical axis direction of the second elastic heat radiating members **802a** and **802b** is applied only to the fixed holder **290** and the fixed base plate **250**. In other words, this load is not applied to the movable base holder **220**, and therefore does not impede the image stabilization.

[0042] The second connecting member **270b** is disposed on the side opposite to the connection surfaces with both ends (fixed portions **801c** and **801d**) of the first heat radiating member **801** of the fixed holder **290** constituting the fixed unit **200b**. This prevents the first heat radiating member **801** and the second connecting member **270b** from coming into contact with each other when the movable base holder **220** moves, thereby preventing the respective flexibility from being impeded.

[0043] FIG. 5B is a sectional view of the main parts in a case where the second elastic heat radiating members **802a** and **802b** are not provided, unlike FIG. 5A. The fixed holder **290** has elastic parts **290a** and **290b**, and both ends of the first heat radiating member **801** are connected to the elastic parts **290a** and **290b**. The elastic parts **290a** and **290b** press (contact) both ends (fixed parts **801c** and **801d**) of the first heat radiating member **801** against (contact with) the heat radiating plate **700**. At this time, the elastic parts **290a** and **290b** have elasticity (flexibility) in the optical axis direction (direction orthogonal to the moving direction of the movable unit **200a**), and suppress the reaction force in the optical axis direction to the image stabilizing unit **200**. The elastic parts **290a** and **290b** may be leaf springs integrated with the fixed holder **290**, or elastic members such as rubber or urethane separated from the fixed holder **290** may be pasted.

[0044] Referring now to FIGS. 6A and 6B, a description will be given of the arrangement of the second elastic heat radiating members **802a** and **802b** and the third elastic heat radiating members **803a**, **803b**, and **803c** provided on the heat radiating plate **700**. FIG. 6A is an arrangement diagram of the second elastic heat radiating members **802a** and **802b** provided on the heat radiating plate **700** when the image pickup apparatus **10** is viewed from the front side (object side). FIG. 6B is an arrangement diagram of the third elastic heat radiating members **803a**, **803b**, and **803c** provided on the heat radiating plate **700** when the image pickup apparatus **10** is viewed from the rear side (rear side).

[0045] The second elastic heat radiating members **802a** and **802b** are arranged on the left and right sides of the image pickup apparatus **10**, respectively. As described above, the first connecting member **270a** is wired between the two second elastic heat radiating members **802a** and **802b** so as to transmit an electrical signal from the image sensor substrate **231** in the vertical direction (transmission direction) of the image pickup apparatus **10**. In FIG. 6B, the transmission direction of the electrical signal is indicated by arrows.

[0046] As illustrated in FIG. 6B, the third elastic heat radiating members **803a** and **803b** are respectively disposed above and below the second elastic heat radiating member **802a**, and the third elastic heat radiating member **803c** is disposed above the second elastic heat radiating member **802b**. Thus, the second elastic heat radiating member **802a** and the third elastic heat radiating members **803a** and **803b** are arranged along a direction parallel to the transmission direction of the first connecting member **270a**. The second elastic heat radiating member **802b** and the third elastic heat radiating member **803c** are disposed along a direction parallel to the

transmission direction of the first connecting member **270a**.

[0047] Referring now to FIG. 7, the control substrate **100** will be described in detail. FIG. 7 is a schematic diagram of the control substrate **100** when viewed from the rear of the image pickup apparatus **10**. As described above, the control IC **101** and the connectors **102**, **103**, and **104** are mounted on the rear side of the control substrate **100**. As illustrated by the dotted lines, control ICs **105a**, **105b**, **105c**, **105d**, and **105e**, which are different from the control IC **101**, are arranged on the front side of the control substrate **100**.

[0048] The control ICs **105a**, **105b**, **105c**, **105d**, and **105e** and the third elastic heat radiating members **803a**, **803b**, and **803c** provided on the heat radiating plate **700** are approximately aligned on the optical axis P. In other words, the control ICs **105a** and **105b** and the third elastic heat radiating member **803a** at least partially overlap each other when viewed from the optical axis direction. Similarly, when viewed in the optical axis direction, the control ICs **105c** and **105d** and the third elastic heat radiating member **803b** at least partially overlap each other. When viewed in the optical axis direction, the control IC **105e** and the third elastic heat radiating member **803c** at least partially overlap each other. This arrangement allows heat from the control ICs **105a**, **105b**, **105c**, **105d**, and **105e** to be transferred (dissipated) to the heat radiating plate **700**.

[0049] In this embodiment, the openings **701a**, **701b**, **701c**, and **701d** are formed on the upper and lower sides of the second elastic heat radiating members **802a** and **802b** of the heat radiating plate **700**. This structure prevents heat from the control ICs **105a**, **105b**, **105c**, **105d**, and **105e** from flowing back to the image sensor substrate **231** via the second elastic heat radiating members **802a** and **802b**.

[0050] Each of the openings **701a**, **701b**, **701c**, and **701d** of the heat radiating plate **700** has a long shape (rectangular shape) in a direction orthogonal to the transmission direction of the first connecting member **270a** (horizontal direction in FIG. 6B), and is configured not to impair the electromagnetic shielding property of the electrical signal. In other words, the length (first length) of the openings **701a** to **701d** in the direction orthogonal to the signal transmission direction in the first connecting member **270a** is larger than the length (second length) of the openings **701a** to **701d** in the transmission direction.

[0051] The third elastic heat radiating members **803a**, **803b**, and **803c** are arranged parallel to the transmission direction of the first connecting member **270a** relative to the second elastic heat radiating members **802a** and **802b**. Therefore, due to the openings that are long in the direction orthogonal to the transmission direction, the backflow of heat between the first and second elastic heat radiating members can be effectively suppressed.

[0052] This embodiment can provide an electronic apparatus with high heat dissipation properties while suppressing the increase in load caused by the displacement of the heat radiating members.

[0053] In this embodiment, an image pickup apparatus is an example of an electronic apparatus, but this embodiment is not limited to this example, and the electronic apparatus may be any electronic apparatus as long as a movable member supported in a displaceable state by a supporting member and a control substrate are connected by a connecting member.

[0054] FIG. 8 is an exploded perspective view of an EVF unit (electronic apparatus) **80** according to this embodiment. The EVF unit **80** includes an EVF panel unit **82** and a lens unit **84**. The EVF panel unit **82** has a diopter adjusting mechanism (not illustrated) and is displaceable (movable) in the Z-axis direction (optical axis direction) relative to the lens unit **84**. The EVF panel unit **82** includes an EVF panel (electronic component) **81** and a panel holder **83** that holds the EVF panel **81**. The EVF panel **81** is adhered to and fixed to the panel holder **83**. Thus, this embodiment is applicable to an electronic apparatus that includes the EVF panel **81** as an electronic component.

[0055] While the disclosure has described example embodiments, it is to be understood that the disclosure is not limited to the example embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0056] This embodiment can provide an electronic apparatus with high heat dissipation property while suppressing an increase in load caused by the displacement of a heat radiating member.
[0057] This application claims priority to Japanese Patent Application No. 2024-037274, which was filed on Mar. 11, 2024, and which is hereby incorporated by reference herein in its entirety.

Claims

1. An electronic apparatus comprising: a fixed unit; a movable unit that holds an image sensor, is opposite to the fixed unit, and is movable relative to the fixed unit; a heat radiating plate opposite to a surface of the fixed unit, which is opposite to a surface of the fixed unit that is opposite to the movable unit; a first heat radiating member that includes a first connector connected to the movable unit, and a second connector connected to a side of the fixed unit, which is opposite to the heat radiating plate; and a second heat radiating member disposed between the second connector of the first heat radiating member and the heat radiating plate, wherein the first heat radiating member and the heat radiating plate are thermally connected via the second heat radiating member.
2. The electronic apparatus according to claim 1, wherein the second heat radiating member contacts each of the second connector of the first heat radiating member and the heat radiating plate.
3. The electronic apparatus according to claim 1, wherein the second heat radiating member is an elastic member.
4. The electronic apparatus according to claim 1, wherein in a case where the first connector and the second connector are viewed from a direction orthogonal to a moving direction of the movable unit, the second connector at least partially overlaps the first connector.
5. The electronic apparatus according to claim 1, wherein the first heat radiating member is flexible in a moving direction of the movable unit.
6. The electronic apparatus according to claim 1, further comprising a second connecting member electrically connecting the movable unit and a control substrate, wherein the second connecting member is disposed on an opposite side of a connection surface of the fixed unit with the second connector.
7. An electronic apparatus comprising: a fixed unit; a movable unit that holds an electronic component and is movable relative to the fixed unit; a first heat radiating member and a second heat radiating member configured to transfer heat from the electronic component; a control substrate configured to transmit a signal; a heat radiating plate provided between the fixed unit and the control substrate; a third heat radiating member configured to transfer heat from the control substrate; and a first connecting member configured to electrically connect the movable unit and the control substrate, wherein the first heat radiating member or the second heat radiating member and the third heat radiating member are connected to the heat radiating plate, wherein the heat radiating plate has an opening, and wherein a first length of the opening in a direction orthogonal to a transmission direction of the signal is longer than a second length of the opening in the transmission direction in the first connecting member.
8. The electronic apparatus according to claim 7, wherein the second heat radiating member and the third heat radiating member are arranged along a direction parallel to the transmission direction of the first connecting member and are connected to the heat radiating plate.
9. The electronic apparatus according to claim 7, wherein the electronic component is an image sensor configured to convert an optical image of an object into an electrical signal, and wherein the movable unit is movable relative to the fixed unit in a direction different from an optical axis direction of an imaging optical system for image stabilization.
10. The electronic apparatus according to claim 9, wherein the direction different from the optical axis direction is a direction orthogonal to the optical axis direction.
11. The electronic apparatus according to claim 9, wherein the fixed unit is configured to adjust a

position of the image sensor in the optical axis direction.

12. The electronic apparatus according to claim 9, wherein the electrical signal output from the image sensor is transmitted to the control substrate.
